

Department of Computer Science and Engineering
University of Dhaka

Buyora

Functional and Non-Functional Requirements

Pre-Publish Buyer Reaction Simulator for Social Commerce

Submitted by

Sumaiya Rahman Soma (SK-07)
H. M. Mehedi Hasan (FH-13)
Md. Abu Bakar Siddique (FH-47)
Chowdhury Shafahid Rahman (AE-55)

Course: CSE 4113 - Internet Programming Lab

Session: 4th Year, 1st Semester

Submitted to

Mr. Abu Ahmed Ferdaus
Md. Ahasanul Alam
Dr. Ismat Rahman

1. System Overview

Buyora is a web application that helps sellers evaluate a promotional product post before publishing or promoting it. The seller provides product and post information, and the system generates different AI buyer personas to simulate how different potential audience groups may react. The results help the seller compare audiences, identify concerns and missing information, and improve the post before promotion.

Note: Buyer reactions are AI-simulated responses. They are not actual customer feedback and should not be treated as guaranteed predictions of real-world buyer behavior.

2. Functional Requirements

ID	Requirement	System Functionality
FR-01	User Authentication	Users can register, log in, log out, recover passwords, and securely access their own account data.
FR-02	Product Management	Sellers can create, view, edit, duplicate, archive, and delete products with price, category, description, features, warranty, delivery, return information, and product images.
FR-03	Promotional Post Management	Sellers can create and edit promotional posts containing headline, body text, offer, call-to-action, hashtags, platform, and related product information. Revised posts can be stored as separate versions.
FR-04	Audience and Simulation Setup	Sellers can provide optional audience information such as location, age range, interests, purchasing preferences, budget sensitivity, and product needs, and select the personas to be evaluated.
FR-05	AI Buyer Persona Generation	The system generates multiple distinct buyer personas representing different potential audience groups based on the product, post, and audience information.
FR-06	Persona Review and Customization	Sellers can review, edit, add, remove, or regenerate buyer personas before starting the simulation.
FR-07	Buyer Persona Evaluation	Each selected persona independently evaluates the same product post according to its assigned characteristics, preferences, needs, and budget sensitivity.
FR-08	Simulated Buyer Reaction Generation	Each persona produces a simulated reaction containing interest, relevance, value perception, price fit, trust, clarity, purchase intention, positive points, questions, and a short buyer-perspective response.
FR-09	Audience Analysis and Comparison	The system calculates consistent comparison scores, ranks personas, and shows which simulated audiences appear more relevant, interested, price-compatible, trusting, or skeptical.
FR-10	Content Issue Identification	The system identifies recurring concerns, objections, buyer questions, and missing or unclear information such as price, warranty, delivery, compatibility, specifications, return policy, or payment details.
FR-11	Post Improvement Suggestions	The system provides clear, prioritized suggestions for improving the promotional content based on persona reactions and identified issues.

ID	Requirement	System Functionality
FR-12	Revised Post Generation	The system can generate an improved version of the promotional post using selected recommendations while keeping the original post unchanged.
FR-13	Result Presentation	Results are presented in an understandable dashboard containing an overall summary, audience ranking, persona reactions, scores, concerns, objections, missing information, and suggestions.
FR-14	Simulation History and Retry	Previous simulations are stored for later review, and failed persona evaluations can be retried without repeating successful evaluations.
FR-15	Simulation Comparison and Report Export	Users can compare two completed simulations or post versions and export or print the analysis as a report.
FR-16	Language Support and Simulation Transparency	The system supports English and Bangla output and clearly states that generated reactions, rankings, and recommendations are AI-simulated rather than actual customer feedback.

3. Non-Functional Requirements

ID	Quality Requirement	Expected Standard
NFR-01	Usability	The interface shall be simple, clear, and understandable for sellers without technical knowledge, with a straightforward step-by-step workflow.
NFR-02	Performance	Product and post operations shall respond promptly, and AI-based analysis shall provide visible progress and complete within a reasonable response time.
NFR-03	Reliability and Recovery	The system shall handle invalid input and partial failures without crashing, preserve completed results, and recover saved simulation data after refresh or service restart.
NFR-04	Security	User accounts, product data, promotional content, and analysis results shall be protected through secure authentication, authorization, encrypted communication, and protected credentials.
NFR-05	Privacy	The system shall avoid collecting unnecessary personal information and shall protect seller-provided content from unauthorized access.
NFR-06	Consistency	All personas in the same simulation shall evaluate the same product and post using a consistent evaluation structure, scoring scale, and output format.
NFR-07	Explainability	Audience rankings, concerns, objections, and recommendations shall include understandable reasons or supporting observations rather than unexplained conclusions.
NFR-08	Maintainability	The application shall use modular and well-structured components so that buyer-persona characteristics, evaluation rules, and system features can be updated or extended easily.
NFR-09	Scalability	The system shall support additional users, products, buyer personas, simulations, languages, and evaluation criteria without major architectural changes.

ID	Quality Requirement	Expected Standard
NFR-10	Accessibility and Compatibility	The application shall be usable on common modern browsers and responsive screen sizes, with readable content, clear labels, keyboard-friendly controls, and accessible result presentation.

4. Feature Flow Summary

Seller Input	Product details, price, promotional post, images, and optional audience information
Buyer Personas	Generate and review different AI buyer personas representing potential audience groups
Simulation	Each persona evaluates the same product post from its assigned buyer perspective
Analysis	Compare audiences and identify reactions, questions, concerns, objections, and missing information
Improvement	Provide suggestions and optionally generate a revised promotional post
Review	View saved results, retry failed evaluations, compare versions, and export the analysis